

Chapter – 7: Factors of Production

For a country like India, the largest contribution to growth and productivity will probably come about from more efficiently using land, labour and capital, thus they must be used more efficiently. — Bibek Debroy, Chairman, Economic Advisory Council to the Prime Minister (2017-24)

Introduction

- Ever wonder – clothes, shoes, school bag, etc. – made
- Every product – production process – before reaching you
- Production process – use resources, inputs – req. – produce – final product
- Resources, inputs – producing goods, services – **factors of production**
 - Ratna – runs – small restaurant – Pause Point – city outskirts
 - Popular – highway travellers – tasty, high-quality food – Pause Point – growing – team – 7 people – assist Ratna – manage business
 - Started 5 yrs. ago – choose location – organise money – rent, equipment – hire staff – buy ingredients – plan – convert dream – success
- **Business** – various inputs, factors – create goods, services – also generate opp. – people – engage – economic activities
- Economics – inputs – used – production process – **factors of production** – classified – 4 types
- Tech. – facilitator – crucial factor – enables business – produce – more goods – same, fewer inputs

Factors of Production

Land (natural resources)

- Word – land – economic – geographical land – BUT also – natural resources – soil, forests, water, air, sunlight, minerals, oil, natural gas
- Recollect – chapter – *Natural Resources and Their Use* – discussed – diff. types – resources – nature gifts us – how to use
- Business – purchase land – OR – pay rent – use it – time period

Labour (human resources)

- Labour – essential – production – phy., mental effort
- Carpenters, farmers, workers, teachers, doctors – varying levels – phy. strength, knowledge, **skill**
- Everyone – contribute diff. – their work – help create – goods, services – society's req.
- **People as a resource**
 - Human beings – key role – economic act. – production process – apply – knowledge, skills, decision-making abilities – create goods services
 - Ex. – police officer – law, order – scientist – new tech. – chef – new recipes
 - All req. – special set – knowledge, skill – perform well
 - Of course – all – dedicated to work – good job
 - Word – **labour** – refers – phy., mental effort – production
 - BUT – **human capital** – refers – specialized skills, knowledge, abilities, expertise – perform labour - human capital – efforts – labour – BUT also – quality, efficiency – labour
- **Facilitators of human capital**
 - **Education and training**
 - Education – help – knowledge – start – basic literacy – extend expertise – specific fields

- Learn in school – enrich knowledge – prepare you – solve real-world problems
- Ex. – civil engineering student – learns principles – design, materials – applied – build infra. – roads, bridges
- Challenge – create – durable, cost-efficient, eco-friendly solu.
- Achieved – **training** – obs. – construction site, testing materials – understand – safety, hands-on application
- Education, training – individuals – prepared – excel – carriers
- **Healthcare**
 - Good health – support development – allow children – regular attendance – learn better
 - Workers – give best – physically, mentally – sound body
 - Do more – shorter time periods – be creative – do not be away – ill health
- **Social and cultural influences**
 - Culture – hard work – cont. improvement – help countries – move forward
 - Japanese concept – *kaizen* – cont. improvement
 - This concept – applied – Japan – mid-1940s – help Japan – higher std. – living – people
 - Another ex. – German work ethic – deep roots – history
 - Germany – renowned – high-quality industrial output – high value – punctuality – attention – detail, quality
 - These qualities – human capital – contribute – their rise – global leader – tech., manufacturing
- **Challenges to human capital**
 - Our nation – long way – various aspects – human capital – independence
 - Literacy – imp. characteristic – population – help enhance – skills, **productivity** – human capital
 - **Adult Literacy rate** – India – 85% – males – 70% – females – 2023 – World Bank estimates
 - Progress – many areas – BUT – India – challenges – develop human capital
 - Economic Survey, India – 2024 – 65% people – below 35 yrs.
 - India – young, productive population – help country – reap benefits – demographic dividend
 - Demographic dividend – refers – benefit – country – large no. – young, working people
 - More people – working, earning – fewer people – depend on them – country – grow business – improve – living std.
 - Take adv. – this potential – individuals – access – quality education, health, training, skills – contribute – nation's progress
 - Today – vast variety – jobs – req. – diff. skills
 - BUT also – India – rich legacy – skill-based knowledge sys., workmanship
 - **India's ancient skill heritage**
 - Ancient Indians – work – form – express – nature, striving – perfection – offering – deity, receiver
 - Tools – worshipped – tradition continue – *Vishwakarma Puja, Ayudha Puja*
 - Creating products – unique blend – *kala* (art), *vidya* (knowledge)
 - Knowledge – pass on – gen. to gen.
 - *Shilpa shastras* – ancient texts – detail design guidelines – sculptures, paintings, buildings, wooden items, jewellery
 - Ex. – texts – sculpture – exact specifications – postures, colours, measurements, proportions
 - **Stitched shipbuilding**
 - Indians – unique stitching tech. – dating – 2000 yrs. – make – ships, boats – conduct – maritime trade – cultural exchange – Indian Ocean
 - Tech. – stitching wooden planks – cords – no nails – flexible – easy navigation

Capital

- Business – req. **capital** – monetary resources – durable assets – machinery, tools, equipment, vehicles, vending carts, comp., shops, factories, office buildings – day-to-day operations
- Ratna – req. money – land lease – furniture, kitchen equipment
- These all – capital – money + human-made resources – produce – goods, services
- Capital – essential – manufacture unit – service sector enterprise
- BUT – where – get capital?
- Personal savings, family, friends – 1st src. – funds, support – individuals – start business – like Ratna
- BUT – Ratna – funds insufficient – loan – bank – meet shortfall
- Paid **interest** – along – part – loan – time period
- Large companies – req. – money – lots – expand business – raise money – gen. public – stock market
- **Stock market** – special market – shares – bought, sold
- Large companies – capital – public – offer them – share – profits – **dividend**

Entrepreneurship

- Start – own business – create – something new – solve problem
- Entrepreneur – person – some up – idea – gather factors of production – make **startup** idea – successful
- Vision – solve problems – innovative products, services – market – benefit – society, nation
- Same time – create job opp. – support livelihoods
- Return – deep sense – satisfaction – dreams – reality – serve people
- **The man who dreamed big for India**
 - **T.R.D. Tata – entrepreneur, industrialist, philanthropist**
 - Jhangir Ratanji Dadabhoy (J.R.D.) Tata – India's greatest entrepreneur – big role – build modern India
 - Born – 1904 – head – Tata Group – largest business grp. – country
 - He believed – business – make money – BUT also – help society
 - Start – India's 1st airlines – Tata Airlines – 1932 – later – Air India
 - His leadership – Tata Group – expand – many areas – steel, cars, power, chem.
 - J.R.D. Tata – also known – care – employees – believed – provide – good working conditions
 - Man – vision, hard work, honesty – 1992 – Bharat Ratna – India's highest civilian award – great service nation

Technology: An Enabler of Production

- Tech. – application – scientific knowledge
- Ex. – camera – convert – light – elec. signals – create – digital image
- Production act. – some form – tech.
- Some early forms – tech. – existed – ancient times – in use today
- Today – newer, adv. – tech. developments – various areas – make lives – easier
- Ex. – payments – 1 click – UPI (Unified Payments Interface) – farmers – adv. weather updates – GPS (Global Positioning Sys.) – shortest routes
- Old tech. – replaced by – new, better
- This process – easier – people, business – things done – improve work
- Ex. – instead – post letters – use email – communicate – quick, low cost
- BUT – tech. progress – not always – replace old tech. – pulleys, wheelbarrows – still used
- Look – ex. – tech. – helping students – learn, build, find – skills, jobs

- **Technology paving the way for accessing knowledge, skills, job opportunities**
 - Variety – online courses – students – govt. platforms – **SWAYAM** (Study Webs of Active Learning for Young Aspiring Minds) – offer courses – Grade 9 onwards – operates – Massive Open Online Courses (MOOCs) – learners – explore sub. – robotics, aquaculture, textile printing – free of cost
 - Students – benefits – own pace – anywhere – pursue – other jobs, courses
 - Online portals – govt.'s **National Career Service** – help people – job opp. various sectors
 - Tech. – eliminate – geographical barrier – allow people – access – knowledge, skills, jobs – India, abroad
 - Services – online portals – links –
 - <https://swayam.gov.in/>
 - <https://www.ncs.gov.in/>

How are the Factors Connected?

- Factors – land, labour, capital, entrepreneurship, tech. – combine – produce – goods, services – proportion – each factor – depend on product
- Ex. – output – agriculture, construction, handicraft sectors – labour-intensive – BUT – semiconductor chips, satellites – more capital-intensive
- Factors – complement each other – interconnected – in case – missing, misused factors – production – inefficient, halted
- Some cases – new tech. – change proportion – output
- Ex. – increase machine use – agriculture – lower dependence – labour
- 3-D printing – revive – dying art forms – textiles – produce handloom products – large scales – serve market
- Production inputs – diff. geographical locations – business – procure them – combine inputs – produce goods, services
- Geographic interconnection – access – varied inputs
- BUT – production act. – severe challenge – supply chain
- **Supply chain** – network – individual, org., resources, act., tech. – production, sale – goods
- Disruption – supply chain – relying src. – far-off places – instead – local inputs – result – halt – production – COVID-19
- Human effort – every stage – production – design, supervise, improve – products, processes
- Ex. – teams – software, electrical, mech. engg. – project managers – sue expertise – develop product
- Entrepreneur – provide guidance – use resources
- Procure resources – land, factory, space, machinery, skilled workers – req. – financial resources
- Together – all inputs – essential – work – puzzle pieces – create – goods, services

Responsibilities towards Factors of Production

- Produce goods, use natural resources – land, water, minerals
- These resources – limited – harmed – not careful
- Ex. – Tamil Nadu – many people – earn money – leather factories
- This help – local economy – BUT – waster – factories – pollute rivers, soil
- Same way – old smartphones – not recycled – lead, mercury – leak – ground, water
- This pollution – dangerous – people, animals, plants

- Imp. – producers – use resources responsibly – meet needs – today – do not make it harder – future gen. – meet theirs
- Try – reduce waste – avoid pollution – protect environment – make products
- Imp. – producers – adopt – sustainable practices – replenish resources – future use
- Business – also responsible – workers, employees – summarized –
 - **Fair compensations and working conditions** – employers – ensure – workers – paid fairly – labour – work – safe env.
 - **Skill development and training** – responsibility – invest – training, education – ensure – workers develop skills – remain competitive – labour market
 - **Workplace right and protections** – laws, regulations – workers' rights – fair treatment – prevent discrimination – provide benefits – health care, paid leave
- Business – motivated – address – social, environmental concerns – their operations – benefit society, biodiversity – Corporate Social Responsibility
- Include – reduce pollution – address well-being – local communities – treat employees, customers – respect